User Input Requirements / NL Prompt (e.g., 'Planetary Rover domain with battery & samples')

Phase 1: Knowledge Elicitation (nodes/elicitation.py)

Node 1: Elicitation Engine

LLM Pydantic Extraction

Structured Pydantic AST Schemas
- DomainBlueprint JSON
- ProblemBlueprint JSON

Phase 2: Code Formulation (nodes/generation.py)

Node 2: Option B Compiler (compile_pddl_programmatically)

Deterministic Compilation

Compiled Lisp PDDL Text Strings
- domain.pddl
- problem.pddl

Phase 3: Symbolic Oracle Validation (graph.py)

Oracle 1: VAL Validator (run_val_validator)

Syntax Valid

Oracle 2: Fast Downward Planner (run_fast_downward)

Plan Found & Solved

Node 4: Maintenance Engine

Syntax / Type Error

Unsolvable / No Plan

Run Historical Test Cases

Phase 4: Domain Maintenance (nodes/maintenance.py)

Historical Suite Passed?

Regression Breakage

All Tests Passed

Update Blueprint in Memory

Capture Raw Error Log in state

Phase 3.2: Fault Triage & Localization (nodes/triage.py)

Node 4: Triage Engine

Log Parsing & Pattern Matching

DiagnosticReport Objects
- ErrorCategory: SYNTAX / TYPE / SOLVABILITY
- RepairOperationType: PRECONDITION_RELAXATION, etc.
- Faulty Element Name & Suspiciousness Rank Score

Route Gatekeeper (route_after_triage)

Iter < Max & Errors Exist

Max Iterations Reached 5/5

Errors Resolved & Reports Empty

Phase 5: Targeted Model Repair (nodes/repair.py)

Node 5: Repair Engine

Inject Triage Hint & AST Blueprint

Reasoning LLM AST Repair

⚠️ Terminate: Max Iterations Limit

✅ SUCCESS: Model Fully Verified

Auto-Save Output to Disk Streamlit_output_file/'domain_name'/domain_N.pddl

Streamlit UI Display (app.py)
- Side-by-Side Before vs. After PDDL Tabs
- Expandable Clickable Diagnostic Cards